PRAVEEN V 241901082

Exercise 2

Website, Email, WHOIS, DNS and Network Footprinting

AIM

To collect information about a target through website, email, WHOIS, DNS and network footprinting techniques, in order to identify the web technologies, email routing details, domain ownership, DNS records, IP addresses and the network path of the target.

PROCEDURE

Part A: Website Footprinting

1. Open a web browser and visit the target website (testphp.vulnweb.com).
2. Use Wappalyzer or BuiltWith to identify the CMS, web server, programming language and frameworks.
3. View the page source (Ctrl+U) to find comments, scripts and hidden links.
4. Open Netcraft or whatweb to find server details: whatweb testphp.vulnweb.com
5. Check testphp.vulnweb.com/robots.txt to find restricted directories.
6. Use HTTrack or the Wayback Machine to view the site structure **and archived pages.**

Part B: Email Footprinting

1. Use theHarvester to collect email addresses of the domain: theHarvester -d testphp.vulnweb.com -b google
2. Use an email tracking tool (eMailTrackerPro or mailtrack) to send a tracked email.
3. Open the received email header and analyze it to find the sender IP, mail server path and timestamps.

4. Use mxtoolbox.com Email Header Analyzer to view the routing details.

Part C: WHOIS Footprinting

1. In Kali Linux, open the terminal and run:
   whois testphp.vulnweb.com
2. Note the registrar, creation date, expiry date, name servers and registrant details.
3. Verify the same details using who.is or ICANN Lookup.

Part D: DNS Footprinting

1. Use nslookup to find the IP address:
   nslookup testphp.vulnweb.com
2. Query specific record types:
   nslookup -type=MX testphp.vulnweb.com
   nslookup -type=NS testphp.vulnweb.com
   nslookup -type=TXT testphp.vulnweb.com
3. Use dig for detailed DNS information:
   dig testphp.vulnweb.com ANY
4. Use dnsenum or dnsrecon to enumerate subdomains:
   dnsenum testphp.vulnweb.com
5. Attempt a zone transfer:
   dig axfr testphp.vulnweb.com @ns1.example.com

Part E: Network Footprinting

1. Use ping to check whether the host is alive:
   ping -c 4 testphp.vulnweb.com
2. Use traceroute to find the network path:
   traceroute testphp.vulnweb.com
3. Use Nmap to discover live hosts and open ports:
   nmap -sn testphp.vulnweb.com
   nmap -A testphp.vulnweb.com
4. Record all the results in the observation table.

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OBSERVATION TABLE

S.No	Footprinting Type	Tool / Command	Information Gathered
1	Website	Wappalyzer / whatweb	Web server, programming language, technologies used
2	Website	robots.txt, page source	Restricted directories, hidden links, comments
3	Email	theHarvester	Email addresses and hostnames of the domain
4	Email	Email header analysis	Sender IP, mail server path, timestamps
5	WHOIS	whois	Registrar, creation and expiry date, name servers
6	DNS	nslookup	IP address and MX, NS, TXT records
7	DNS	dig / dnsenum	Detailed DNS records and subdomains
8	Network	ping	Host is alive, response time, TTL value
9	Network	traceroute	Hops and routers between source and target
10	Network	nmap	Live host, open ports, services, OS details

OUTPUT

whatweb output:

http://testphp.vulnweb.com [200 OK] Country[UNITED STATES], HTTPServer[nginx/1.19.0], PHP[5.6.40], IP[44.228.249.3]

whois output:

Domain Name: VULNWEB.COM

Registrar: Eurodns S.A.

Creation Date: 2010-06-14

Name Server: NS1.EURODNS.COM

nslookup output:

Name: testphp.vulnweb.com

Address: 44.228.249.3

ping output:

4 packets transmitted, 4 received, 0% packet loss

traceroute output:

1 192.168.1.1 1.2 ms

2 10.10.0.1 5.4 ms

3 * * *

...

10 44.228.249.3 210 ms

nmap output:

PORT STATE SERVICE VERSION

80/tcp open http nginx 1.19.0

Host is up.

RESULT

Thus, website, email, WHOIS, DNS and network footprinting were successfully performed on the target. Information such as web technologies, email addresses, domain registration details, DNS records, IP address, network path, open ports and running services was collected.